EEG1001 AGV project: Team working and Design Presentation

AGV Project: Team working is central!

- All students will be allocated to teams of 3 or 4
  - Team membership, including email addresses, will be shared via loop
  - Use email to arrange initial team contacts/meeting
- Discuss and agree on communication methods (phone, social media, meetups etc.) for ongoing co-ordination and collaboration within your team

Team working: some starting points

- Collectively discuss and agree all aspects of team "ground rules":
  - how to address workload allocation, responsibilities, roles, expectations
  - how to address any problems in team interactions
  - revisit these ground rules on an ongoing basis!
- Practice professional, constructive, interpersonal interactions at all times
  - mutual respect for *diversity*
  - ensure space to *listen* to contrary views
- Refer to the Team Working- Resources uploaded to the loop

8 Core Design Principles of Prosociality

1. Strong team identity and understanding of purpose
2. Fair distribution of contributions and benefits
3. Fair and inclusive decision-making
4. Monitoring of agreed behaviours
5. Graduated responding to helpful and unhelpful behaviour
6. Fast and fair conflict resolution
7. Authority to self-govern
8. Appropriate relations with other people and groups

Review and discuss this short explanatory video!

Team Presentations: Preliminary AGV Design

- Collectively discuss, agree and develop *preliminary* design aspects of your AGV (challenges: design& communication)
  1. Overall chassis dimensions and outline component layout
  2. Drive Mechanism & calculations as per the loop document
  3. Steering Mechanisms & Calculations as per the Loop document
- 10% of total project mark (6.5% of overall module mark)
  - **Absent/non-participating students will receive 0%**

Team Presentations: Preliminary AGV Design

- Objective of presentation is to evaluate:
  - Teamwork
  - Each team's initial *progress* on model AGV design
  - Communication skills
  - Self-directed learning

Team Presentations: Scheduling

- Presentation schedule will be posted on loop at end of week 9
- Presentations will take place over weeks 10-12
- 15-minute slots:
- 8 minute presentation + 7 minute Q&A
- All team members must be *present* and must *contribute*
- After joint presentation, team will be asked questions
  - May be collective or individual!
- Turn around of teams will be swift
  - Make sure you are all present and ready on time
- You only have to attend *one* presentation (your own team)

Presentation Structure and Style

- Background guidance on giving presentations (via loop):
  - Preparing a Presentation (Dublin Region Higher Education Alliance)
  - The 4Ps of Giving a Good Presentation (University of Hull)
  - Review and discuss within your teams!

Refer: https://loop.dcu.ie/course/view.php?id=62015&section=6#

Team Presentations: Software

- Use google slides (via your DCU google suite account) to prepare your team presentation slide deck
- Work together on a **single, shared google slides file** (stored in one nominated team member's google drive)
- You will be required to submit both a google slides *link* and a PDF copy of your slides in advance
- In your scheduled slot, you will present *directly* from google slides

Sample slides suggestions- Title slide

EEG1001 Model AGV Preliminary Design

Team Name: (e.g. “Mario Kart”…)

Team Number: M-3

Presentation Date: DD:MM:YY

Team member	Contribution
Alice Perry	Steering
William Dargan	Component layout/Chassis
Anne Butler	Drive
Margaret Hamilton	Teamwork co-ordination-slide preparation

Sample slide content: Chassis layout

- Include *sketch(es)* showing preliminary overall chassis and component layout
- 2D plan + elevation view sketches are sufficient (no need for 3D/isometric views)
- Use hand (pencil) sketches that are then photographed/scanned
- Check for good contrast!

Sample slide content: Drive Design

- Include *sketch(es)* (hand drawn/pencil) and *calculations*
- Sketches should show what the constituent components of the equations refer to on the drive mechanism, e.g. N_s number of teeth on the spur gear, n_s is revolutions per minute (RPM) of the spur gear
- 2D sketches are sufficient (no need for 3D/isometric views)
- Calculations:
  - Present in a step by step manner per pages 5 and 6 of Car Driving and Steering Mechanisms.
  - **Do *not* use handwritten presentation for *calculations*: format properly in google slides (including appropriate mathematical symbols, Greek characters, sub-/superscripts etc.).**
  - Beware of false precision!

Sample slide content: Steering Design

- Include *sketch(es)* and *calculations*
- Sketches should show what the constituent components of the equations refer to on the AGV: e.g. wheelbase **L**, track **W**, turning angles **δ**, etc.
- 2D plan view hand/pencil sketches are sufficient (no need for 3D/isometric views)
- Calculations should be presented in a step by step manner per pages 18 and 19 of Car Driving and Steering Mechanisms

Getting to know google slides

- Study/practice google slides tutorial materials:
- What can you do with Slides? (via google help centre)
- Google Slides Tutorial (via youtube)
- How to Use Google Slides ("Ultimate Tutorial Guide")
- … and many more!
- Suggest different team members review different resources and then share and discuss the techniques and approaches they have encountered